Vidyalankar Institute of Technology
Semester VII – Information Technology - Mid Semester Assessment – 1

Date: 17/08/2026	Natural Language Processing	PC_PEC/ PE-6	20 Marks/45 mins.

			CO
1	**Solve any two (5 marks each)**		
	A	i) What is the basic difference between stemming and lemmatization? For the words “studies”, “running” — show the output after (a) stemming and (b) lemmatization in a table, and mark where the two differ. ii) Classify the following as inflectional or derivational morphology, with justification: “creative", “boxes”	CO1
	B	What is Language Ambiguity? Identify the type of ambiguity in following statements and explain the reason of ambiguity. i) Flying planes can be dangerous. ii) Can you open the window? iii) She saw a bat. iv) John told Mike that he had won the prize.	CO2
	C	Using the following corpus, calculate the probability of the sentence **<s> Alice and Ravi played at the park </s>** using a Bigram Language Model (without Laplace smoothing). Corpus: <s> the park was crowded </s> <s> Ravi and Alice went to the cinema </s> <s> Alice and Ravi played at the park </s> <s> Ravi went to the park </s> <s> Alice and Ravi were happy </s>	CO2
2	**Solve anyone (10 marks each)**		
	A	Explain finite state morphology as a model for morphological analysis, and construct a simple finite state transducer (FST) diagram that accepts the inflected forms of the regular and irregular verbs like walked, walks, walking, cut, catch, speak, eat, etc.	CO2
	B	Find the probability of the sentence: “astronomers saw stars with ears” using CKY algorithm and the given PCFG. S → NP VP 1.0 VP → V NP 0.7 VP → VP PP 0.3 PP → P NP 1.0 P → with 1.0 V → saw 1.0 NP → NP PP 0.4 NP → astronomers 0.1 NP → ears 0.18 NP → saw 0.04 NP → stars 0.18 NP → telescope 0.1	CO3

CO1	Recall fundamental concepts of NLP such as stages of language processing, grammar, and basic text preprocessing techniques like tokenization and stemming.
CO2	Explain morphological, syntactic, and semantic components of language using appropriate models and tools.
CO3	Apply parsing algorithms and semantic techniques (like WSD and discourse resolution) to analyze linguistic structures.